ELECTRON MICROSCOPY FACILITY

C-WING, GROUND FLOOR, GC-04, DIVISION OF BIOLOGICAL SCIENCES
INDIAN INSTITUTE OF SCIENCE, BANGALORE-560012

NEGATIVE STAIN TEM SAMPLE PREPARATION

Workstation Location: Room GC-04 (Specimen Prep Area)	**Equipment:** Glow Discharge Unit
Pre-requisite: Purified protein/nanoparticle sample in solution form (0.01–0.1 mg/mL)	**Target Grid Type:** Carbon-coated TEM Grid (300/400 Mesh)

PRECAUTION & HEAVY METAL SAFETY GUIDELINES: Heavy metal stains such as Uranyl Acetate (UA) and Formate are toxic and light-sensitive. Always handle stains using protective gloves and eye protection. Dispose of all stain-contaminated tips, filter paper, and waste in heavy-metal hazardous waste containers.

PREPARATION & STAINING PROTOCOL

1. **Grid Hydrophilization (Glow Discharge):** Place carbon-coated TEM grids onto a clean glass slide with the carbon side facing upward. Subject the grids to glow discharge treatment (air plasma, 15–30 seconds) to render the hydrophobic carbon film hydrophilic for optimal sample adsorption.
2. **Droplet Array Setup:** On a strip of clean Parafilm, aliquot droplets in sequence: one 3.5–5.0 µL drop of sample solution, two 10–20 µL drops of Milli-Q water (for washing), and two 10 µL drops of 1–2% Uranyl Acetate solution.
3. **Sample Adsorption:** Hold the glow-discharged grid gently at the edge using anti-capillary tweezers. Place the grid carbon-side down onto the sample droplet for 30–60 seconds to allow target particles to adsorb.
4. **Washing Steps:** Pick up the grid and gently touch the edge with Whatman No. 1 filter paper to blot away excess liquid. Touch the carbon face onto the first and second Milli-Q water droplets sequentially, blotting between steps to remove salt buffers.
5. **Staining & Blotting:** Touch the grid face onto the first stain droplet for 5 seconds and blot immediately. Then, place it onto the second stain droplet for 30–45 seconds. Carefully blot remaining stain from the edge, leaving a thin, uniform liquid film to air-dry.
6. **Drying & Grid Storage:** Allow the grid to air-dry completely in a covered petri dish or grid box for at least 10 minutes. Store prepared grids in a designated TEM grid box, record specimen details, and sign the facility logbook before loading into the microscope.

Note: Negative staining provides high-contrast structural screening for biological macromolecules and nanoparticles prior to high-resolution cryo-EM or TEM analysis. Clean handling, optimal stain thickness, and thorough buffer washing are essential to avoid stain precipitation and structural artifacts.